

Diesel Engines**V 396 TE 94****Maintenance Schedule****M050383/20E**

MTU Motoren- und Turbinen-Union Friedrichshafen GmbH
88040 Friedrichshafen / Germany
Telephone (07541) 90-0
Telex 734 280-0 mt d
Telefax (07541) 90-3928

Low operating and maintenance costs as well as operational reliability and availability depend on maintenance and servicing being carried out in compliance with our specifications and instructions.

The overall system, of which the engine is an integral part, must be maintained in such a way as to ensure trouble-free engine operation at all times. For this purpose always:

- ensure that sufficient fuel is available,
- ensure that the combustion air is dry and clean,
- use only dry, clean compressed air,
- use only clean, filtered raw water.

Moreover, it is essential that:

- maintenance tasks be completed by trained personnel,
- suitable tools be used,
- genuine spare parts and fluids and lubricants as per the current MTU Fluids and Lubricants Specification A001061 be used.

Our Product Support Service will always be available should assistance be required.

Preventive maintenance instructions.

- Special care should be taken to keep the machinery plant in a clean and serviceable condition at all times to facilitate early detection of possible leaks and prevent subsequent damage.
- Protect rubber and synthetic components from oil and fuel, never treat with organic detergents. Wipe with a dry cloth only.
- Always replace all seals and gaskets.

MTU Maintenance Concept

The maintenance concept for MTU products is a preventive maintenance concept and features the maintenance echelons W1 to W6 as outlined below.

Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability.

Maintenance Echelon W1: Operational Checks.

Maintenance Echelons W2, W3 and W4: Periodic maintenance tasks to be performed during out-of-service periods without the need for engine disassembly.

Maintenance Echelon W5: This echelon requires partial engine disassembly.

Maintenance Echelon W6: Major overhaul. This echelon requires complete engine disassembly.

The time intervals according to which the W2 to W6 Maintenance Echelons and the relevant checks and tasks involved are to be completed are based on operational experience. They have been determined so as to ensure correct engine operation until the next scheduled maintenance echelon.

Use of a maintenance predictor can mean changes to the operating hours given in the Maintenance Frequency Chart.

Specific operational conditions may require modification of the Maintenance Schedule.

Note:

The codes on the following pages refer to the groups and subgroups of the Task Descriptions in Section G of the Engine Operation Manual.

This Maintenance Schedule may refer to parts which are not installed on your engine, these may be disregarded.

Application Group

1DS: Fast vessels with low load factors

Load profile

Load referenced to fuel stop power %	100	70	< 10
Time element %:	10	70	20

Maintenance Frequency Chart

Maintenance Echelon	W1: Operational checks, daily	X
	W2: Operating hours	250
	Limit, months	6
	W3: Operating hours	500
	Limit, year	1
	W4: Operating hours	1 500
	Limit, year(s)	2
	W5: Operating hours	3 000
	Limit, year(s)	6
	W6: Operating hours	6 000
	Limit, year(s)	18

Maintenance Task Schedule -Overall System-

See page 4

Maintenance Task Schedule -Engine-

Code No.	One-time operations after the first 50 operating hours. - With a new engine, after W5 maintenance echelon, or after W6 major overhaul	
G00.049	Attachments	Check tightness of securing screws and nuts
G06.101	Valve gear	Check valve clearances, adjust if necessary
G12.311	Fuel pre-filter	Clean
G12.311	Fuel pre-filter	Replace filter elements
G12.321	Fuel duplex filter	Replace filter elements
G13.112	Engine coolant pump	Check relief bore for obstructions
G13.123	Raw water pump	Check relief bore for obstructions
G13.511	Bilge pump	Check relief bore for obstructions
G14.019	Engine coolant system	Clean strainers in coolant return lines
G16.002	Engine oil	Take sample and analyse
G16.121	Engine oil edge-type filter	Operate filter handle several times

Code No.	Maintenance Echelon W1: - Operational checks	
G10.051	Exhaust system	Check exhaust gas colour Drain condensate (if drain valve provided)
G10.181	Intercooler	Check drain line for water discharge and obstructions
G10.211	Air filter	Check contamination indicator
G10.401	TC control	Drain condensate from actuating cylinder
G12.311	Fuel pre-filter	Operate filter handle several times Drain water and contaminants
G12.311	Fuel pre-filter	Check differential pressure
G14.011	Engine coolant	Check level
G16.002	Engine oil	Check level
G84.001	Monitoring system	Carry out lamp test
G84.002	Engine operation	Check running noises, Check engine and external pipework for leaks, Check speeds, pressures and temperatures - if gauges are provided
G86.051	Compressed air system	Check operating pressure Drain condensate